



**ADDRESSING INFORMAL SKILLED SERVICE
DISCOVERY THROUGH A TRUSTED TASK-MATCHING
PLATFORM IN KIGALI**

BSc. in Software Engineering

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List of Acronyms / Abbreviations

API - Application Programming Interface

eKash - Rwanda national interoperable payment switch

ERD - Entity Relationship Diagram

KYC - Know Your Customer (identity verification)

MoMo - Mobile Money (e.g. MTN Mobile Money)

MVP - Minimum Viable Product

NIDA - National Identification Agency (Rwanda)

SDG - Sustainable Development Goal

SME - Small and Medium-sized Enterprise

SUS - System Usability Scale

TaPa - Task Pairing (project name; here: TaPa Trust)

UI - User Interface

UML - Unified Modeling Language

UTAUT - Unified Theory of Acceptance and Use of Technology

UX - User Experience

Chapter One: Introduction

1.1 Introduction and Background

Urban residents, households, and small businesses regularly require short skilled services to keep daily life and work running: minor electrical repairs, plumbing support, furniture assembly, moving or lifting help, deep cleaning, event setup, cooking support, and basic technical setup. Although individually small, delays in finding a reliable worker disrupt household routines, office operations, and business activity.

Kigali, Rwanda's main urban and business centre, is a suitable setting because it concentrates households, tenants, offices, small and medium-sized enterprises (SMEs), students, and informal service providers. Rwanda also has a very large informal employment sector: the National Institute of Statistics of Rwanda reported about 3,269,809 employed persons in the informal sector in 2023, roughly 82.5% of total employment (NISR, 2024). Many workers therefore depend on informal income-generating activity and personal networks, which makes skilled-service discovery a shared concern for both those who need work done and those who do the work. This connects to a recognised global challenge captured in Sustainable Development Goal 8 — productive employment and decent work for all — and to evidence that digital labour platforms are expanding in developing countries while workers still lack stable client access and social protection (Datta et al., 2023). Rwanda's national development strategy similarly emphasises technology, innovation, and services as engines of inclusive growth (World Bank, 2022).

Traditionally, people find skilled workers through relatives, neighbours, house helps, office assistants, WhatsApp groups, social-media recommendations, or word-of-mouth referrals. These methods are familiar and low-cost, but they are unreliable when a task needs a specific skill, tool, location, or urgent response, and they reveal little about a worker's skill level, reliability, or previous work. The research literature shows that digital systems can improve service access by structuring search, communication, ratings, reviews, and feedback (Dellarocas, 2003; Resnick et al., 2000; Tadelis, 2016).

Rwanda's digital environment supports such software: RURA reported active mobile-cellular subscriptions of about 13.3 million as of early 2025 (RURA, 2025). This study treats that figure

with appropriate caution — mobile-cellular subscriptions are not the same as smartphone ownership or affordable mobile-data access — and therefore designs for a lightweight, low-bandwidth, phone-first experience rather than assuming high-end devices.

It is important to state plainly that several platforms already help people find workers in and beyond Kigali — including Taskerz, Kazispace, Akazi, MESEE, and the government-backed Kora Job Portal, alongside global models such as TaskRabbit and Airtasker. This project does not claim that worker-discovery platforms are absent. Its contribution lies elsewhere, in the trust infrastructure that sits behind discovery, which the competitor analysis in Chapter Two examines explicitly rather than assuming a gap.

The problem also has two distinct sides that need different design treatment. On the demand side, requesters need to evaluate whether a worker is skilled, available, and trustworthy before inviting them into a home, office, or business. On the supply side, informal skilled workers need a way to make their skills, reliability, work history, and earnings visible so they can be found and re-engaged beyond their personal network. TaPa Trust is therefore framed as a two-sided trust platform, and the design implications of each side are carried through the objectives, research questions, and evaluation.

Preliminary survey findings from 25 respondents (treated as indicative rather than statistically representative, given the small convenience sample) support the relevance of the problem: 23 respondents had previously needed someone for a short skilled task, and 17 rated their likelihood of using a trusted app for verified short skilled services as 4 or 5 out of 5. The most requested categories were plumbing, cleaning, moving or lifting, cooking or catering, electrical help, furniture assembly, and mounting or installation. Respondents valued verified identity, ratings and reviews, clear skill profiles, task-completion history, clear pricing, and — most strongly — the ability to rebook the same trusted worker (19 of 25).

This project therefore proposes **TaPa Trust**, a trust-centered worker profile and rebooking system for informal skilled services in Kigali. Unlike general task-matching platforms, TaPa Trust focuses on the trust infrastructure behind informal skilled-service discovery. The system helps users evaluate workers through skill profiles, verification tiers, ratings, completed-task

history, mutual check-in/check-out records, payment-status tracking, dispute reporting, safety check-ins, and preferred-worker rebooking, and it helps informal skilled workers build visible digital records of their skills, work history, ratings, and earnings.

1.2 Problem Statement

Individuals, households, and small businesses in Kigali frequently need short skilled services — minor electrical repairs, plumbing, furniture assembly, moving or lifting, deep cleaning, event setup, cooking support, and basic technical assistance — yet finding reliable workers is informal and fragmented. Most users depend on relatives, neighbours, WhatsApp groups, or unverified referrals. These methods do not reliably surface workers with the right skills, tools, location, availability, or reliability, which creates a service-discovery gap that is acute when help is urgent or specialised (Graham et al., 2017; NISR, 2024).

A second problem is the lack of trust, verification, and accountability. Informal methods rarely offer structured ways to verify identity, assess skills, compare previous work, confirm that a task was actually completed, agree on how long it took, or resolve complaints. Research on digital marketplaces shows that trust mechanisms — ratings, reviews, reputation history, and feedback — reduce uncertainty between requesters and providers (Dellarocas, 2003; Resnick et al., 2000; Tadelis, 2016), but the same work warns that these mechanisms can be inflated, manipulated, or biased against newcomers, and so must be designed around those weaknesses rather than assumed sufficient.

A third problem is limited worker visibility and difficulty maintaining repeat relationships. Skilled informal workers often have real experience but depend on word-of-mouth, which limits their ability to show a track record, display skills, receive structured feedback, or be rebooked. The literature on platform economies notes a related risk — disintermediation — where requesters who find a good worker simply contact them directly next time, bypassing the platform. This is the same behaviour that the survey's strongest signal points to (19 of 25 would be more likely to use a system that lets them save and rebook a trusted worker). TaPa Trust treats that repeat-relationship behaviour as a feature to support rather than leakage to prevent.

To address these problems, this work introduces TaPa Trust, a trust-centered worker profile and rebooking system for informal skilled services in Kigali. The proposed solution will include

worker profiles, skill categories, verification tiers, admin-reviewed identity and credential verification, ratings and reviews, completed-task history, mutual check-in/check-out, payment-status tracking, dispute reporting, safety check-ins, worker earnings history, and preferred-worker rebooking. These features aim to reduce uncertainty in informal skilled-service discovery by helping users evaluate worker credibility while helping skilled workers build visible records of reliability and completed work.

Central contribution. The contribution of TaPa Trust is not any single feature — most are individually present on mature global platforms — but the design and local evaluation of a closed trust-accountability loop: verified identity → agreed and recorded time → mutual completion confirmation → payment status, in which each step validates the next and the whole chain is anchored to repeat (rebooking) relationships. No platform currently operating in Kigali is shown to close this full loop for short, practical, physical tasks.

1.3 Project's Main Objective

To design, develop, and evaluate TaPa Trust, a trust-centered worker profile and rebooking system that improves how users verify, and re-engage informal skilled workers in Kigali.

1.3.1 List of the Specific Objectives

The overall aim is broken down into the following specific objectives, each framed to be specific, measurable, achievable, relevant, and time-limited (SMART):

1. To investigate trust, verification, worker-visibility, payment-confidence, safety, dispute-handling, and rebooking challenges in informal skilled-service discovery, through a literature review, competitor analysis, the preliminary survey (25 respondents), and 8–12 user interviews, during Weeks 1–3.
2. To design and develop a functional TaPa Trust prototype implementing a complete trust-accountability loop — role-based worker and requester profiles, skill-based worker profiles with ratings and task history, mutual check-in/check-out, mutual completion confirmation with simulated payment-status tracking, and preferred-worker rebooking — by Week 6. Additional trust features (verification tiers, dispute escalation, a worker earnings dashboard, and safety check-ins) will be implemented incrementally as time

permits; any not implemented will be fully specified as design contributions and future work (see 1.5.1, Feature Prioritization).

3. To evaluate the prototype through task-based testing with 8–15 participants drawn from both service requesters and informal skilled workers, measuring usability (targeting a mean SUS score of at least 75), task-completion rate (target of at least 80%), perceived usefulness, perceived trust (using an adapted McKnight trust scale), ease of evaluating a worker, and ease of rebooking a worker, during Weeks 7–8.

1.4 Research Questions

1. What trust and accountability challenges do users face when evaluating informal skilled workers in Kigali?
2. What challenges do informal skilled workers face in making their skills, reliability, work history, and earnings visible to potential clients?
3. Which digital trust mechanisms can improve confidence in informal skilled-service discovery and repeat engagement?
4. How usable, useful, and trust-supporting is the TaPa Trust prototype in helping users evaluate and rebook informal skilled workers?

1.5 Project Scope

Geographical Scope

The study focuses on Kigali, with no claim to national generalisation. Prototype testing will draw on accessible areas of Kigali, particularly Nyamirambo (a concentration of informal workers, small businesses, and households — the supply side) and Kimihurura (offices, restaurants, institutions, and businesses — the demand side). Using both areas lets the study consider supply and demand together.

Content Scope

The project covers short skilled practical services complete within a limited time: minor electrical support, plumbing, furniture assembly, deep cleaning, moving or lifting, event setup,

cooking or catering, mounting or installation, and basic technical setup — the categories most selected in the preliminary survey.

Technical Scope

The system is a functional prototype rather than a commercial product. Core features implement the trust loop described in 1.3. The prototype will also include simulated payment-status tracking and admin-reviewed verification workflows. It will not process real money or connect directly to NIDA, MTN MoMo, eKash, or Smile ID during the capstone phase; these external integrations are designed and specified as future work.

Language Scope

Because many informal workers in areas such as Nyamirambo are primarily Kinyarwanda speakers, UI language is a design consideration rather than an afterthought. The prototype's core flows (profile, check-in/out, completion, rebooking) will use clear, simple labels suitable for Kinyarwanda localisation; the capstone prototype will be delivered in English / Kinyarwanda and full multilingual localisation is noted as future work. Stating this demonstrates demographic sensitivity to both sides of the platform.

User and Testing Scope

Problem validation draws on the 25-respondent preliminary survey; further data collection involves 8–12 interviews and prototype testing with about 8–15 participants. Testing will include both service requesters and informal skilled workers, since the system is a two-sided trust platform. Testing focuses on whether users can complete the trust loop — evaluate a worker, record check-in/out, confirm completion, and indicate willingness to rebook.

Exclusion Scope

Real mobile-money escrow, NIDA-linked identity verification, Smile ID integration, insurance, large construction, long-term employment contracts, government-document handling, and emergency-response functionality are excluded from the MVP. The prototype will simulate the relevant workflows through status tracking, admin review, and safety-check design, so the logic can be demonstrated and evaluated without live financial or national-ID systems.

Time Scope

The project is conducted within the capstone period of approximately eight weeks. The schedule is deliberately front-loaded: problem investigation and design occupy the early weeks, the Tier 1 build runs through the middle of the period, and prototype testing and evaluation are reserved for Weeks 7–8. A buffer is placed before testing so that any development delay does not erode the evaluation phase, which is where the core data is generated. The full week-by-week breakdown is presented in the Research Timeline (Section 1.8, Table 3).

1.5.1 Feature Prioritization (Tiered Scope)

To keep the build realistic within the capstone timeline while still aiming for the full vision, features are organised into three tiers. The project commits firmly to Tier 1, which constitutes the contribution; Tiers 2 and 3 are implemented incrementally as time allows, and anything not built is specified as future work. Work proceeds strictly top-down: no new feature is started until the current one is demonstration-complete, so that at every stopping point a coherent, working system exists.

Table 1: Feature Prioritization (Tiered Scope)

Tier	Features	Rationale
Tier 1 — MVP (must complete)	Role-based requester & worker profiles + auth; worker profiles with skills, ratings, task history; mutual check-in/check-out; mutual completion confirmation + simulated payment-status; preferred-worker rebooking	These five close the trust loop end-to-end (identity → time → completion → payment status → rebook) and form a complete, defensible system on their own. Check-in/out is included here because it is the project's sharpest original mechanism, resolving the common informal dispute over how long a job took.
Tier 2 — High value (if time allows)	Verification tiers (Unverified / Peer-Verified / Admin-Certified, badged);	Strengthen the trust signal, add an escalation path tied to payment-status holds, and create a portable income

	dispute reporting with admin review; worker earnings dashboard	record for workers (SDG 8). Admin-reviewed; no external APIs required.
Tier 3 — Stretch / specified future work	Safety check-in (notify-a-trusted-contact); earnings charts/visualisation; real integrations (NIDA / Smile ID, MoMo escrow, eKash)	Valuable but heavier or dependent on external systems. The safety feature is a lightweight notify-a-contact layer, explicitly not an emergency-response system. Integrations remain designed and specified, forming the post-capstone roadmap.

1.6 Significance and Justification

This study addresses a practical, validated service-access problem: 23 of 25 respondents had previously needed a worker for a short skilled task, and 17 rated their likelihood of using a trusted app at 4 or 5 out of 5. If implemented, TaPa Trust can reduce the delay, uncertainty, and reliance on scattered referrals that users face when they need reliable help quickly.

The project also benefits informal skilled workers by helping them build visible digital service records. Instead of depending only on word-of-mouth referrals, workers can display skills, completed tasks, ratings, verification status, and earnings history, which may support repeat clients and stronger income visibility — a direct, supply-side mechanism aligned with the decent-work and inclusive-growth aims of SDG 8 (Datta et al., 2023).

Academically, the study contributes to software engineering by demonstrating how trust infrastructure can be designed for informal service platforms through verification tiers, check-in/check-out records, payment-status tracking, safety check-ins, dispute workflows, and rebooking mechanisms. Its distinctive contribution is the closed trust-accountability loop — identity, time, completion, and payment status linked so each step enforces the next — designed for, and evaluated in, the Kigali informal context, an African urban setting where such localised evidence is limited (Gashema & Alain, 2022).

1.7 Research Budget

Table 2: Research Budget

Item	Purpose	Cost estimate (RWF)
Internet / data bundles	Literature review, survey, development research, and prototype testing	20,000
Transport / communication	Reaching participants in Nyamirambo & Kimihurura for interviews and testing	30,000
Figma / design tools	UI/UX design and prototyping using free or student tiers	0
Hosting / backend services	Prototype deployment, database testing, and backend setup	35,000
Participant incentives / data support	Small mobile-data support for interview and testing participants.	20,000
Contingency	Unexpected project costs	15,000
Total estimated budget		120,000

1.8 Research Timeline

The timeline below reflects the heavier, tiered build. Tier 1 scaffolding (authentication and data model) begins in Week 4 rather than Week 5, and a deliberate buffer is placed before testing so that any development delay does not eat into the evaluation phase, which is where the core data comes from.

Table 3: Research Timeline

Activity	Wk 1–2	Wk 3–4	Wk 5–6	Wk 7–8
Literature review & competitor analysis	■			
Survey collection & interviews	■	■		
Requirements analysis		■		

UI/UX design & system diagrams		■		
Tier 1 scaffolding (auth + data model)		■		
Tier 1 build (trust loop) to completion		■	■	
Buffer + Tier 2 features (as time allows)			■	
Prototype testing & feedback				■
Data analysis & evaluation				■
Report writing & final submission			■	■

Chapter Two: Literature Review

2.1 Introduction

This chapter reviews literature and existing systems relevant to trusted skilled-service discovery, with attention to digital labour platforms, two-sided marketplaces, trust and reputation systems, payment assurance and task-lifecycle accountability, worker visibility and platform ethics, on-demand micro-services, and technology adoption. The aim is to establish what existing studies and systems already do, what weaknesses remain, and where TaPa Trust contributes. Rather than treating each source in isolation, the review is organised thematically so that recurring findings, tensions, and gaps can be synthesised across the literature and then connected to the design of TaPa Trust.

The review followed a thematic approach. An initial search identified a broad pool of candidate sources, from which the most relevant were selected on the basis of their treatment of digital service matching, trust and reputation mechanisms, informal-worker visibility, payment and task-lifecycle accountability, technology acceptance, and prototype evaluation. Sources were located using keywords such as “digital labour platforms,” “online service marketplace,” “trust in online platforms,” “reputation systems,” “informal employment,” “gig economy,” “escrow and payment assurance,” “technology acceptance,” and “on-demand service applications.” Literature was drawn from scholarly databases and indexes (such as Google Scholar, IEEE Xplore, the ACM Digital Library, ScienceDirect, and SpringerLink) together with institutional and statistical reports from organisations including NISR, the ILO, the World Bank, and RURA. Preference was given to peer-reviewed and original sources, and to recent work where the field is evolving quickly, while seminal works were retained for foundational concepts. The chapter is organised into an existing-system analysis (2.2), a thematic review of related work (2.3), a synthesis of the reviewed literature (2.4), an assessment of the strengths and weaknesses of existing systems (2.5), and a concluding statement of the research gap (2.6).

2.2 Overview of Existing System

Several systems address parts of the problem, but they differ from TaPa Trust in focus, depth of trust infrastructure, or local fit. The table below includes the platforms that overlap most directly

with this work — including those omitted from the earlier draft — so the comparison is explicit rather than assumed.

Table 4: Summary of Existing Systems

Existing system	Main focus	Limitation for this study	TaPa Trust comparison
Taskerz	Live Kigali platform for short services (repairs, moving, mounting, cleaning) with MoMo payment and 'verified taskers'	Directly overlaps with task discovery and booking; depth of trust mechanisms (e.g. mutual time records, dispute escalation, portable earnings) is not demonstrated	TaPa Trust centres on the trust loop: verification tiers, mutual check-in/out, payment-status accountability, disputes, earnings history, and rebooking
Kazispace	All-in-one platform linking skilled, semi-skilled, and informal workers with employers across sectors	Broader gig/employment platform, not a trust/rebooking system focused on short physical tasks	TaPa Trust focuses narrowly on worker trust records and repeat engagement for short practical tasks
Akazi	Kigali freelancing / worker-profile and hiring platform	More profile- and hiring-oriented; less focused on task-lifecycle accountability	TaPa Trust adds task history, check-in/out, payment status, disputes, and rebooking
MESEE	Rwandan home-services marketplace (cleaning, handymen, plumbing, electrical,	Home-services marketplace; trust depth around mutual time/completion and	TaPa Trust focuses on the closed trust loop and worker-side visibility, not only booking

	repairs) with vetted providers and ratings	portable worker records not demonstrated	
Kora Job Portal	Government-backed (RDB) job-matching platform (launched 2019)	Focuses on employment matching, not short-task trust workflows	TaPa Trust focuses on informal skilled micro-service trust and rebooking
HouseHelp Rwanda	Local domestic-worker matching	Centres on domestic support, not short skilled micro-services	TaPa Trust covers repairs, setup, moving, cleaning, and other practical tasks
Vuba Vuba	Local food/grocery delivery	Delivery-focused; no skilled-worker discovery	TaPa Trust matches human skilled services, not goods delivery
TaskRabbit / Airtasker	Global task marketplaces with profiles, ratings, escrow payment, background checks, insurance	Not localised to Kigali; mature trust stack assumes infrastructure and budgets unavailable here	TaPa Trust adapts trust mechanisms to the local informal context and a low-bandwidth, phone-first design

This comparison shows that TaPa Trust does not claim to be the first platform for finding workers. Its contribution is narrower and deeper: it focuses on the trust infrastructure behind informal skilled-service discovery. While Taskerz, Kazispace, Akazi, MESEE, and Kora support worker discovery or job matching, TaPa Trust focuses on verification tiers, task history, mutual check-in/check-out, payment-status accountability, dispute escalation, safety check-ins, earnings history, and repeat-worker relationships — and on whether, taken together as a closed loop, these measurably improve trust.

2.3 Review of Related Work

2.3.1 Digital Labor and Platform Economies

Digital labour platforms operate as two-sided marketplaces, creating value by linking two distinct groups — service requesters and service providers — and reducing the cost and difficulty of finding, coordinating with, and trusting the other side (Rochet & Tirole, 2003). Two-sided-market theory highlights that such platforms are shaped by cross-side network effects: each side becomes more valuable as the other grows, which means the platform must attract both sides more or less simultaneously. This gives rise to the well-known cold-start, or “chicken-and-egg,” problem: a marketplace is unattractive to requesters when few workers are present, and unattractive to workers when few requesters are present, so early growth is difficult and must be designed for deliberately rather than assumed.

A substantial and increasingly critical body of work cautions that digital and gig platforms do not automatically benefit the workers they connect. They can lower barriers to entry and ease task allocation, but they may also intensify competition, exert algorithmic control, and expose workers to weak social protection and downward pressure on earnings (Graham et al., 2017; Sundararajan, 2016). A recent systematic review of the gig economy confirms that this tension persists across contexts, with platform work expanding rapidly while the distribution of its benefits remains uneven (Alauddin et al., 2024). Evidence focused on developing economies and Sub-Saharan Africa similarly shows that online and platform-mediated work holds real promise for women and youth, yet workers continue to lack stable client access and protection (Datta et al., 2023; International Labour Organization, 2020).

These findings carry two implications for TaPa Trust. First, because the cold-start problem is structural, the platform is scoped narrowly to a clearly defined local need rather than attempting broad coverage, which lowers the threshold of participation required for either side to find value. Second, because efficiency-focused platform design can disadvantage workers, TaPa Trust deliberately emphasises repeat, trust-based relationships and worker visibility — designing with the worker’s interest in mind — rather than optimising solely for transaction volume or price competition.

2.3.2 Trust and Reputation Systems

Trust is a central problem in online marketplaces because users routinely interact with people they do not already know. Reputation systems address this by allowing participants to build, display, and consult a history of past behaviour. Foundational work argues that an effective reputation system depends on entities being long-lived and re-identifiable, on feedback about past interactions being reliably captured and made visible, and on that feedback actually guiding future decisions (Resnick et al., 2000). The digitisation of word-of-mouth through online feedback mechanisms extends informal reputation to strangers at scale, but it also introduces new vulnerabilities that offline reputation did not face (Dellarocas, 2003).

The same literature is markedly critical of reputation mechanisms in practice. Ratings are frequently inflated or sparse; reviewers may withhold honest negative feedback for fear of retaliation; feedback can be strategically manipulated or gamed; and newly joined providers face a cold-start disadvantage because they have accumulated no history and are therefore hard to trust on day one (Dellarocas, 2003; Tadelis, 2016). A reputation score, in other words, is necessary but not sufficient: a system that relies on ratings alone inherits all of these weaknesses. This matters acutely for short skilled services, where a worker physically enters a home, office, or business, so the cost of misplaced trust is high and immediate.

The local evidence reinforces the importance of designing for trust explicitly. A study of e-commerce adoption in Rwanda found that transactions are harder precisely because buyers and sellers do not know one another, and that deliberately increasing customer trust is a viable strategy for driving uptake (Gashema & Alain, 2022). TaPa Trust responds to both the strengths and the documented weaknesses of reputation systems: rather than treating ratings as sufficient, it pairs reviews with identity verification, verification tiers, and a visible, transaction-linked task history, so that a worker's standing rests on corroborated evidence rather than on potentially manipulable star ratings, and so that verified newcomers are not entirely disadvantaged by the absence of prior reviews.

2.3.3 Payment Assurance and Task-Lifecycle Accountability

A recurring weakness of both informal arrangements and lighter digital platforms is that trust tends to collapse around two specific points: money and time. Workers fear performing a job and then not being paid, while requesters fear paying for work that is incomplete or substandard. Critically, in informal skilled work many disputes are not about whether a task was done but

about how long it took, and therefore how much is owed — a disagreement that ratings and reviews, which are recorded only after the fact, cannot resolve. These are predictable, structural failure points rather than incidental problems.

The wider platform literature suggests that addressing these points requires mechanisms that operate during the transaction, not only after it. Escrow-style models, in which funds are committed up front and released only once both parties confirm completion, convert a soft completion signal into a financially enforced commitment and are associated with fewer payment-related disputes. More generally, task-lifecycle accountability — recording an agreed start and finish, requiring mutual confirmation of completion, and tracking an explicit payment status — closes the gap between what was promised and what is later contested. Each step in the lifecycle produces evidence that constrains the next, which is the mechanism by which uncertainty is reduced.

TaPa Trust models this lifecycle directly through mutual check-in/check-out records and simulated payment-status tracking, so that the timing of work is agreed and recorded by both parties and completion is confirmed mutually before payment status advances. Because building and operating real financial integration is beyond the scope and licensing of a capstone prototype, full mobile-money escrow (for example via MTN MoMo or the national eKash switch) is specified as future work; the prototype demonstrates and evaluates the accountability logic without processing real money.

2.3.4 Worker Visibility, Safety, and Platform Ethics

Beyond matching requesters to providers, platforms shape three further outcomes that the trust-and-reputation literature treats as increasingly important: whether informal workers can accumulate a portable record of their work, whether both parties are kept safe during in-person service, and whether personal and identity data are handled responsibly. Because the informal sector accounts for the large majority of employment in Rwanda (NISR, 2024), and because such workers typically depend on personal networks and lack any formal income record, a structured and exportable history of completed, rated, and paid tasks is not a cosmetic feature but a potential contribution to financial inclusion — the kind of visible track record that can support repeat clients and, over time, access to formal services.

Safety is a dimension that ratings alone do not capture. Short skilled services involve a worker entering a requester's home, office, or business, or a worker travelling to an unfamiliar location, and a generally well-rated worker may still pose a situational risk to a particular person in a particular place. This is especially salient for female requesters inviting workers into their homes and for workers attending jobs alone. A lightweight, privacy-respecting safety mechanism — such as sharing a task's location and expected end time with a trusted contact — addresses a real concern that purely reputational systems leave unmet, while remaining distinct from, and not a substitute for, formal emergency services.

Finally, the move from informal referral to a verified platform raises ethical and data-protection obligations. Identity verification, sometimes termed Know Your Customer (KYC), is precisely what makes the platform trustworthy, yet identity data is sensitive and its collection must follow principles of data minimisation, clear and specific consent, secure and access-controlled storage, and limited retention. Adoption research in developing-country platform contexts indicates that perceived risk and privacy concerns shape whether users take up a platform at all, alongside its usefulness and ease of use (Mensah & Khan, 2023). TaPa Trust therefore treats worker visibility, safety, and responsible data handling as first-class design concerns rather than later additions, and the prototype evaluation uses mock or demo identity data so that these commitments are honoured during testing.

2.3.5 On-Demand Micro-Service Platforms

On-demand service platforms demonstrate that small, discrete tasks can be effectively organised through software, typically by supporting a common sequence of discovery, worker selection, communication, task confirmation, and feedback, with mature platforms adding verification, pricing transparency, and customer support to strengthen trust. Emerging-market home-service platforms such as SweepSouth in South Africa and Urban Company in India demonstrate that this model extends beyond the Global North to contexts more comparable to Rwanda (SweepSouth, n.d.; Urban Company, n.d.). This body of practice shows that the basic mechanics of organising micro-services are well understood; what differentiates platforms is less the presence of these features than the depth and local fit of the trust mechanisms around them.

A recurring critique in the literature is that platforms attempting to cover too many service types tend to dilute trust and quality, whereas narrowly scoped platforms sustain trust more effectively

(Sundararajan, 2016; Tadelis, 2016). Breadth increases the difficulty of verifying competence and of building category-specific reputation, while narrow scope allows clearer expectations and more comparable histories. TaPa Trust applies this lesson directly: it stays deliberately narrow — short, practical, physical skilled tasks — and invests instead in depth of trust infrastructure (verification tiers, mutual time and completion records, payment-status accountability, dispute escalation, and rebooking), distinguishing it from delivery platforms, full domestic-placement services, and broad global task marketplaces.

2.3.6 Technology Acceptance and Usability

A platform is only useful if its intended users are willing and able to use it. The Technology Acceptance Model holds that perceived usefulness and perceived ease of use are primary determinants of whether a technology is adopted (Davis, 1989). More recent work on platform adoption in developing countries extends this view using the Unified Theory of Acceptance and Use of Technology (UTAUT2), showing that trust, perceived risk, and privacy concerns also shape uptake of on-demand digital platforms, alongside usefulness and ease of use (Mensah & Khan, 2023). For a trust-centred platform aimed at users who may have limited prior experience with such applications, these factors interact: a system that is trustworthy in principle will still fail if it is hard to use, if users do not believe it solves their problem, or if they are uneasy about how their data is handled.

For this reason, TaPa Trust is not evaluated solely as a technical artefact. Its evaluation, detailed in Chapter Three, combines task-based testing with user feedback and measures usability (via the System Usability Scale), perceived usefulness, perceived trust (via an adapted trust scale), task-completion rate, and the perceived ease of evaluating and rebooking a worker. This multi-dimensional approach reflects the literature’s insight that acceptance is driven jointly by usability and trust, neither of which alone is sufficient.

2.4 Summary of Reviewed Literature

Taken together, the reviewed literature converges on several points. Digital platforms can materially improve service access by structuring search, communication, and reputation, but the value they create is not automatic and is shaped by network effects and by design choices that may help or harm workers. Reputation mechanisms reduce uncertainty yet carry

well-documented weaknesses — inflation, manipulation, retaliation, and cold-start bias — so they must be reinforced rather than relied upon alone. Payment and time accountability are central to trust and are best addressed during the transaction through lifecycle mechanisms, not only through after-the-fact ratings. Worker visibility, safety, and responsible data handling are recurring but under-addressed concerns. And adoption depends jointly on usability and trust. The table below maps each thematic area to its key idea, its relevance to TaPa Trust, and the gap it identifies.

Table 5: Summary of Reviewed Literature

Literature area	Key idea	Relevance to TaPa Trust	Gap identified
Digital labour & two-sided markets	Platforms link two groups; value depends on network effects	TaPa Trust serves both requesters and workers	Cold-start problem; efficiency focus can disadvantage workers
Trust & reputation systems	Ratings, reviews, and history reduce uncertainty	TaPa Trust uses verified profiles, tiers, and task history	Ratings can be inflated/manipulated; newcomers disadvantaged
Payment & task-lifecycle accountability	Money and time are where trust collapses; in-transaction mechanisms help	Mutual check-in/out and payment-status tracking	Lighter platforms leave completion a soft signal; duration disputes unaddressed
Worker visibility, safety & ethics	Portable records, safety, and data protection matter beyond matching	Earnings history, safety check-in, minimised verification data	Rarely combined or localised; safety seldom addressed
On-demand micro-services	Small tasks can be organised in software;	TaPa Trust stays narrow and deep on trust	Broad platforms dilute trust;

	narrow scope sustains trust		delivery/placement differ in kind
Technology acceptance & usability	Usefulness, ease of use, trust, and privacy drive adoption	Evaluated via SUS, usefulness, trust scale, completion rate	Trust-in-principle fails without usability; needs local testing

2.5 Strengths and Weaknesses of Existing Systems

Existing systems address discovery, but not all of them address deeper trust infrastructure. Global task platforms (TaskRabbit, Airtasker) offer mature trust stacks — background checks, escrow, insurance, and dispute services — but are not localised and assume infrastructure, regulatory context, and operating budgets that are unavailable in the Kigali informal context. Local platforms (Taskerz, Kazispace, Akazi, MESEE, Kora) demonstrate that digital service access works in Rwanda and validate the underlying need, but the depth of their trust mechanisms — mutual time records, payment-status accountability tied to completion, portable worker earnings, and structured dispute escalation — is either narrower in scope or not publicly demonstrated. Informal referral methods remain accessible and low-cost and carry a form of personal trust, but they lack verification, comparable history, and any structured accountability. The table below summarises these strengths, weaknesses, and the corresponding TaPa Trust improvement.

Table 6: Strengths and Weaknesses of Existing Systems

Existing system	Strengths	Weaknesses	TaPa Trust improvement
TaskRabbit / Airtasker	Mature model: profiles, categories, ratings, escrow, background checks, insurance	Not localised to Kigali; assumes infrastructure and budgets unavailable here	Localised, low-bandwidth trust mechanisms adapted to the informal context
Taskerz	Live Kigali short-task platform with MoMo	Depth of trust (mutual time, dispute escalation,	Closed trust loop: check-in/out, payment-status

	payment and 'verified taskers'	portable earnings) not demonstrated	accountability, disputes, earnings
Kazispace / Akazi	Connect skilled and informal workers; profiles and hiring	Broad gig/employment focus; less task-lifecycle accountability for short tasks	Narrow focus on short-task trust records and repeat engagement
MESEE	Local home-services marketplace with vetted providers and ratings	Trust depth around mutual time/completion and portable records not demonstrated	Worker-side visibility and the full trust loop, not only booking
Kora Job Portal	State-backed; reach and credibility for job matching	Employment matching, not short-task trust workflows	Informal skilled micro-service trust and rebooking
HouseHelp Rwanda	Locally relevant; domestic-worker matching	Domestic placement, not short skilled micro-services	Broader short skilled categories with trust infrastructure
Informal referrals (WhatsApp, word-of-mouth)	Accessible, low-cost, carry personal trust	No verification, comparable history, or accountability; exclude those without networks	Structured verification, history, and accountability open to all

2.6 General Comment and Conclusion

The gap addressed by this study is not the complete absence of task or worker-discovery platforms; on the contrary, several already operate in and around Kigali (Taskerz, n.d.; Kazispace, n.d.; Metropole Services Exchange [MESEE], n.d.; Akazi, 2020; Rwanda Development Board, 2019). Rather, the gap is the limited localised design and evaluation of trust infrastructure for informal skilled services — including verification tiers, task-lifecycle records, payment-status accountability, safety check-ins, worker earnings history, dispute escalation, and

repeat-worker relationships — and, in particular, whether linking these into a single closed trust-accountability loop measurably improves user trust and willingness to rebook in the Kigali context.

Chapter Three: System Analysis and Design

3.1 Introduction

This chapter presents the research design, development model, data-collection and analysis methods, ethical considerations, the proposed model, system diagrams, architecture, and development tools. The study uses mixed methods to understand user needs, Agile prototyping to guide development, and UML and system modeling to represent the proposed system structure and interactions.

3.2 Research Design Including the Development Model Used

The study uses a mixed-methods design and an Agile, prototype-based development model (Denscombe, 2012). Survey data identify common needs, current discovery methods, trust concerns, and desired features; interviews provide deeper insight into reliability, pricing, availability, and trust. Agile prototyping lets user feedback shape the trust features and is well suited to the tiered build described in 1.5.1.

Sampling Strategy

Purposive and convenience sampling will target relevant participants — individuals, tenants, households, small-business owners, office workers, and informal skilled workers — with testing focused on 8–15 participants from Nyamirambo and Kimihurura, covering both sides of the platform. Findings are treated as indicative, not generalisable.

Data Collection Methods

Data are collected from three sources. The preliminary survey (25 respondents) supports problem validation and identifies common service needs, trust concerns, and desired platform features. Semi-structured interviews (8–12 participants) explore user experiences with informal service discovery, worker reliability, pricing and availability expectations, safety concerns, and expectations for a trusted platform, on both the requester and worker sides. Prototype testing (8–15 participants) gathers task-based performance and feedback on whether users can post a task, evaluate a worker, record and confirm check-in/out, confirm completion, and indicate willingness to rebook.

Data Analysis Methods

Each data source is analysed with an appropriate method. Survey data are analysed using descriptive statistics — frequencies, percentages, and ranked responses — to summarise needs, trust factors, and willingness to use the platform. Interview responses are analysed thematically, identifying recurring themes around trust, reliability, pricing clarity, worker quality, availability, completion, safety, and repeat-worker relationships. Prototype feedback is analysed using usability and trust metrics: task-completion rate, perceived usefulness, perceived trust (adapted McKnight trust scale), user satisfaction, and post-test perceived ease of evaluating and rebooking a worker. The earlier pre/post ‘reduction in difficulty finding workers’ measure is replaced, because a single-session prototype test cannot reliably capture longitudinal behavioural change; a post-test perceived-ease item combined with a trust-perception scale is more appropriate to a single-session evaluation.

Usability Target and Approach

The evaluation targets a mean System Usability Scale (SUS) score of at least 75. A SUS of 70 marks only the lower boundary of acceptable usability; because TaPa Trust's value proposition is trust among users who may be new to task-matching apps, a higher bar is appropriate. The SUS measures perceived ease of use rather than trust or features, so this target is pursued through deliberate interface-design choices, not by adding functionality. The following principles guide the design:

- Short, obvious core flows — minimise the number of steps and decisions needed to post a task, check in and out, confirm completion, and rebook a worker.
- Frictionless trust steps — the novel check-in/out and mutual-completion actions use single, clearly labelled buttons with immediate confirmation, since unfamiliar steps are where first-time users are most likely to hesitate.
- Plain, consistent language and icons, suitable for the target users (see the Language Scope in 1.5), avoiding technical jargon.
- Immediate feedback and clear, recoverable error handling — every action produces a visible response, and mistakes can be undone or corrected.
- First-use guidance — simple empty-state prompts help first-time users complete their first task.

To reach the target in practice, a short pilot test with two to three users will be run before the graded sessions so that the most obvious friction points can be fixed first, and realistic seeded data will be used so the prototype feels populated rather than empty. Testers will receive a brief, consistent introduction to the task without being coached through the interface, so the score reflects the design rather than the facilitation.

Importantly, the SUS result is treated as a finding to be analysed, not merely a threshold to pass. If the mean meets or exceeds 75, the design is reported as having met the usability bar. If it falls short, the shortfall is analysed at the level of individual SUS items to identify which aspects of the interface scored low, and the proposed improvements are reported as part of the evaluation and future work. A care is also taken not to over-simplify away the trust mechanisms that constitute the contribution: the two-party check-in/out and completion steps add a small, deliberate amount of friction, and the evaluation reports whether they were made lightweight enough to preserve usability.

Ethical Considerations

Participation is voluntary and based on informed consent; responses are confidential, anonymised, and used only for academic purposes, subject to institutional/ethics approval before data collection. Because TaPa Trust involves identity verification and stored task history, the design follows data minimisation, clear and separate consent, secure restricted-access storage, and retention only for as long as the evaluation requires. Crucially, the prototype will use mock identity documents or demo data during testing; real national-ID documents will not be collected from participants unless institutional approval, secure storage, and explicit consent procedures are in place. The platform also supports fair ratings, responsible feedback, dispute reporting, and protections against misuse of worker profiles.

3.3 Proposed Model Diagram

The model includes service requesters, skilled workers, and administrators, with a trust layer spanning all three. Requesters post tasks, evaluate and select workers, confirm completion, review, and rebook. Workers create profiles, list skills, request verification, accept tasks, record check-in/out, complete work, and build earnings history. Administrators review verification and credentials, monitor tasks, and handle disputes.

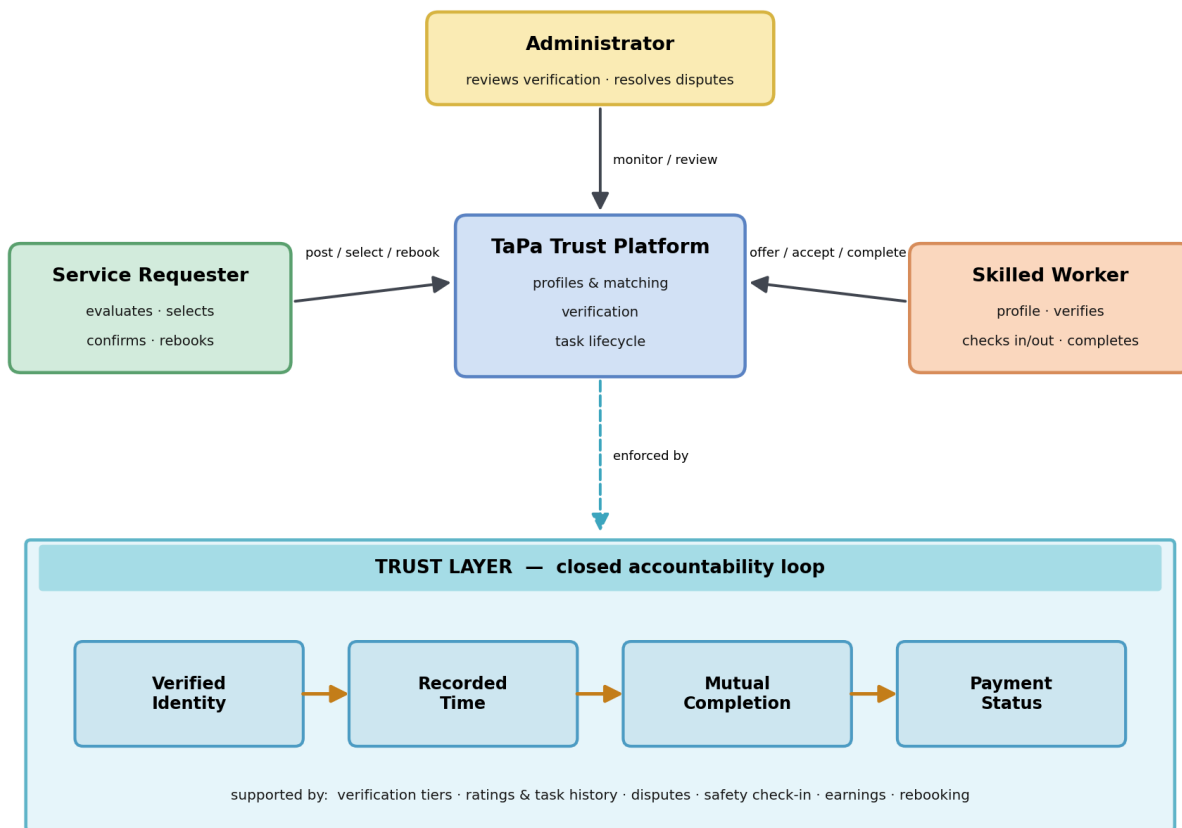


Figure 1: Proposed TaPa Model Diagram

Building on the original class set (User, Worker, SkillCategory, Task, Booking, Review, Notification, Admin, Report, PaymentStatus), the trust-infrastructure pivot adds the following classes/entities:

- **VerificationRequest** — A worker's submitted identity/credential evidence awaiting admin review; carries status and reviewer notes.
- **SkillVerificationTier** — The worker's trust tier (Unverified / Peer-Verified / Admin-Certified) and the basis for it.
- **CheckInRecord** — Mutual start/finish timestamps for a task, each confirmed by both parties; basis for duration and dispute prevention.
- **SafetyContact** — A trusted contact a worker (or requester) can notify with task location and expected end time (Tier 3).

- **EarningsRecord** — Per-task and aggregated earnings for a worker; basis for the exportable income summary.
- **DisputeResolution** — A dispute raised by either party, with evidence, admin ruling, and outcome tied to payment-status hold.
- **SavedWorker** — A requester's saved/preferred worker, enabling one-tap rebooking.
- **PaymentStatus** — Simulated status of payment for a booking (e.g. pending, confirmed, released) linked to completion.

3.4 Class Diagram

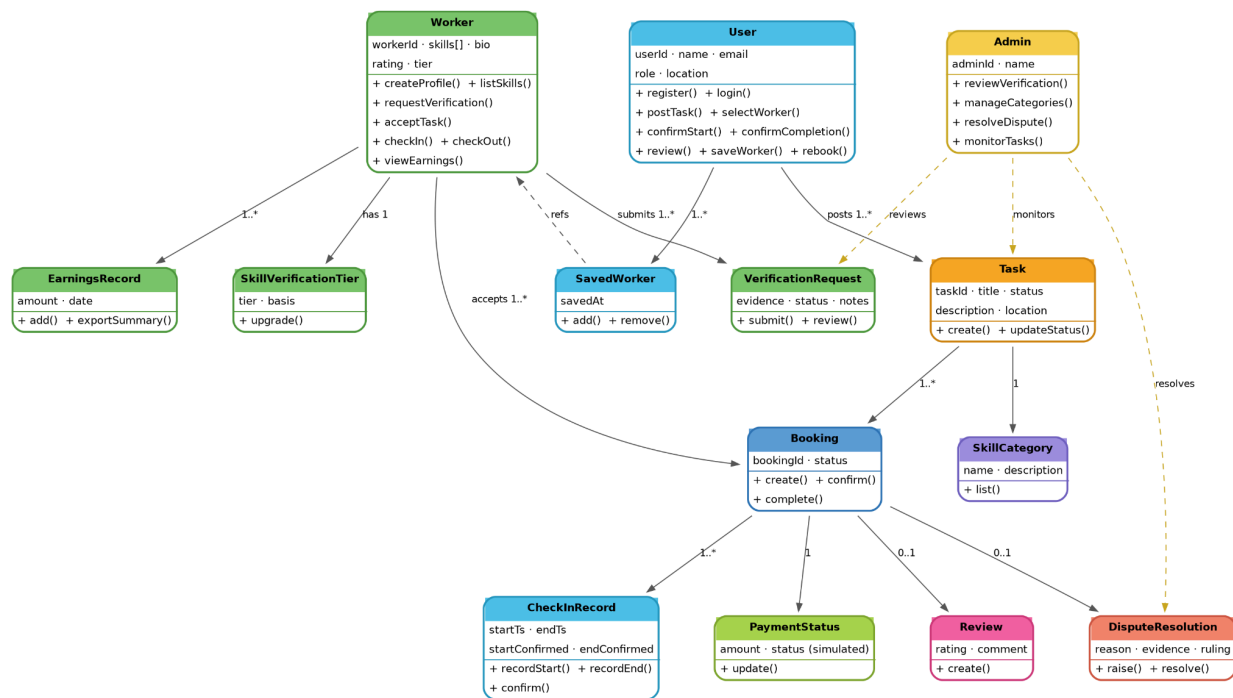


Figure 2: Class Diagram

The class diagram in Figure 2 represents the main classes of TaPa Trust and the relationships among them. The three principal actors are User (the service requester), Worker (the skilled worker), and Admin, supported by Task, Booking, and SkillCategory. A User can post many Tasks, save many Workers, and rebook them; each Task belongs to one SkillCategory and can lead to one or more Bookings, and a Booking links a User, a Task, and a Worker.

Around this core sit the trust-infrastructure classes that distinguish the system. A Worker holds one SkillVerificationTier, submits one or more VerificationRequests (reviewed by an Admin),

and accumulates EarningsRecords. Each Booking can generate one or more CheckInRecords (mutual start and finish timestamps confirmed by both parties), a PaymentStatus that advances only on confirmed completion, an optional Review, and an optional DisputeResolution that an Admin resolves. SavedWorker records a requester's preferred workers to enable one-tap rebooking.

Each class encapsulates its own attributes and methods — for example, Worker exposes requestVerification(), checkIn(), checkOut(), and viewEarnings(), while Booking exposes confirm() and complete(). Taken together, the verification, check-in/out, payment-status, dispute, and rebooking classes form the closed trust-accountability loop that is the project's central contribution, rather than a simple post-and-review marketplace.

3.5 System Architecture

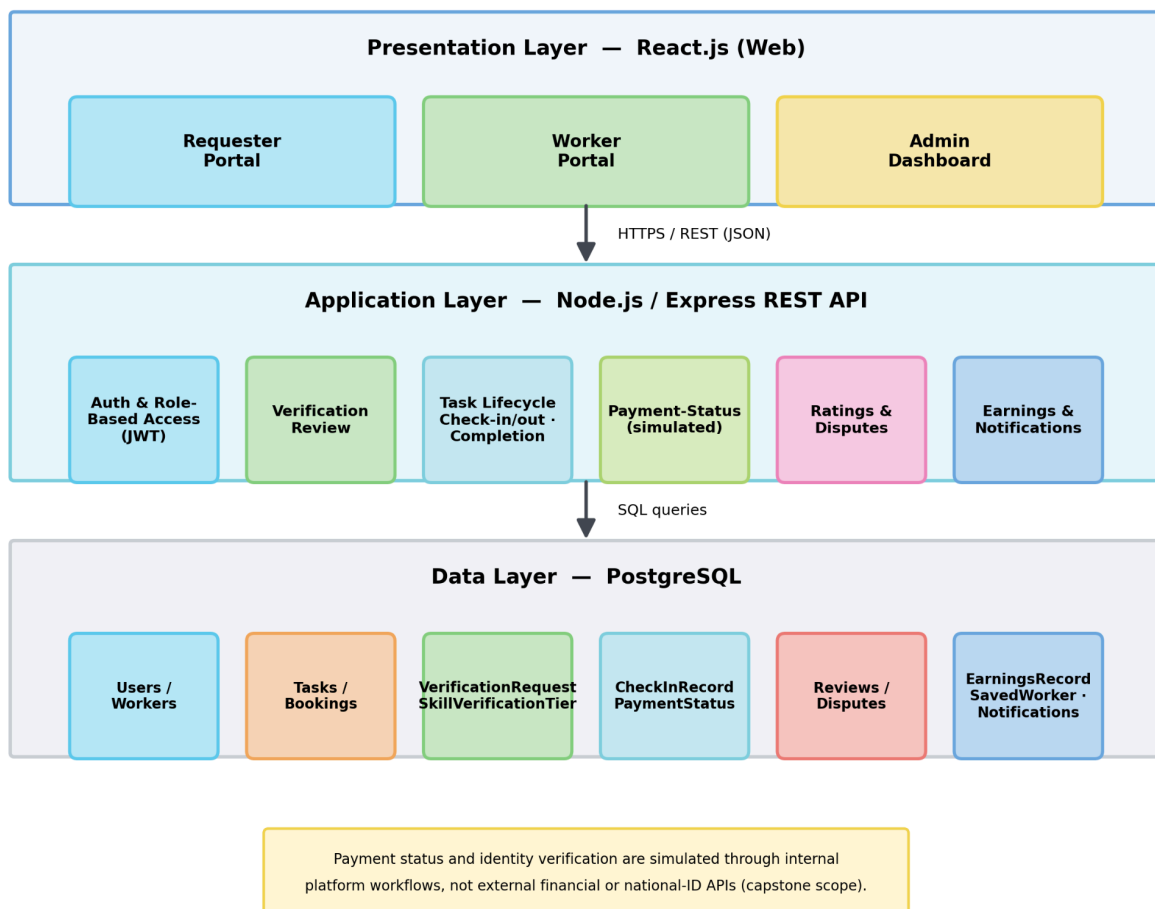


Figure 3: System Architecture Diagram

TaPa Trust follows a three-layer client–server architecture, illustrated in Figure 3. The presentation layer is a React.js web client providing three role-based interfaces: a Requester Portal, a Worker Portal, and an Admin Dashboard. These communicate with the backend over HTTPS using REST (JSON) requests.

The application layer is a Node.js and Express REST API that holds the system's logic. It handles authentication and role-based access control (JWT), verification review, the task lifecycle (check-in/out and completion), payment-status tracking, ratings and disputes, and earnings and notifications. This layer is where the closed trust-accountability loop is enforced, since it coordinates verification, mutual confirmation of time and completion, and the advancement of payment status.

The data layer is a PostgreSQL relational database that persists all entities — users and workers, tasks and bookings, verification requests and skill-verification tiers, check-in records and payment status, reviews and disputes, and earnings, saved workers, and notifications. The application layer reads and writes to this layer through SQL queries.

To keep the project within the scope of a capstone prototype, payment status and identity verification are simulated through internal platform workflows rather than integrated with external financial services or the national identification system. Full mobile-money and national-ID integration is identified as future work, so the architecture demonstrates and evaluates the trust logic without processing real money or handling real identity documents.

3.6 Entity Relationship Diagram

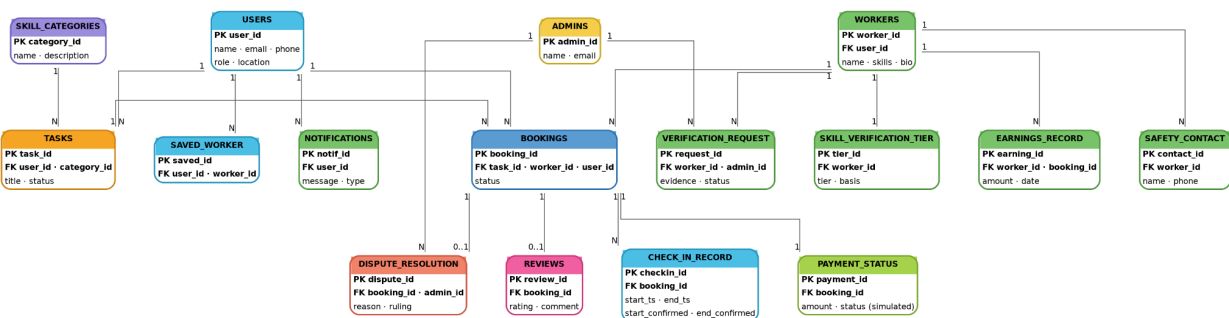


Figure 4: Entity Relationship Diagram

The entity relationship diagram in Figure 4 shows the database structure that supports the system, including the primary key (PK) and foreign key (FK) of each entity and the cardinality of each relationship. The core entities are Users, Workers, Skill Categories, Tasks, and Bookings. A user can post many tasks and save many workers; each task belongs to one skill category; and a booking links one user, one task, and one worker, forming the central record around which accountability is tracked.

Surrounding this core are the trust-infrastructure entities. Each worker has one Skill Verification Tier, can submit many Verification Requests (reviewed by an admin), and accumulates many Earnings Records and Safety Contacts. An Admins entity holds platform administrators; the `admin_id` foreign key on Verification Requests and Dispute Resolutions references it, so every review and ruling is tied to the administrator who performed it. Each booking can have many Check-In Records (mutual start and finish timestamps), exactly one Payment Status (simulated), an optional Review, and an optional Dispute Resolution. Notifications and Saved Workers are linked to users.

The cardinalities follow standard notation: 1 denotes one, N denotes many, and 0..1 denotes an optional single record. This structure ensures that every trust-related action — verification, check-in/out, payment status, review, and dispute — is stored against the specific booking and worker it concerns, which is what makes the accountability loop auditable.

3.7 Use Case Diagram

Figure 5 shows the system's use cases grouped by actors within the TaPa Trust system boundary. The Service Requester can register and log in, post a task, evaluate and select a worker, confirm check-in/out, confirm completion, rate and review, raise a dispute, and save or rebook a worker. The Skilled Worker can register, create a profile and list skills, request verification, accept tasks, record check-in/out, and view earnings. The Administrator can review verification, manage categories, resolve disputes, and monitor tasks. The actor boundaries reflect deliberate design decisions. "Register / Log in" is shared because both requesters and workers are authenticated users of the same platform. The Administrator deliberately has no transactional cases (posting, accepting, or paying for tasks) and is limited to oversight — reviewing verification and resolving disputes — so that the party adjudicating trust is kept separate from the parties transacting, which

protects the integrity of the accountability loop. All interactions are modelled as direct associations between actor and use case; no «include» or «extend» relationships are used, in order to keep the prototype's scope explicit and unambiguous. Collectively, the trust-related cases — verification, mutual check-in/out, completion confirmation, disputes, and rebooking — are what distinguish TaPa Trust from a basic post-and-review marketplace.

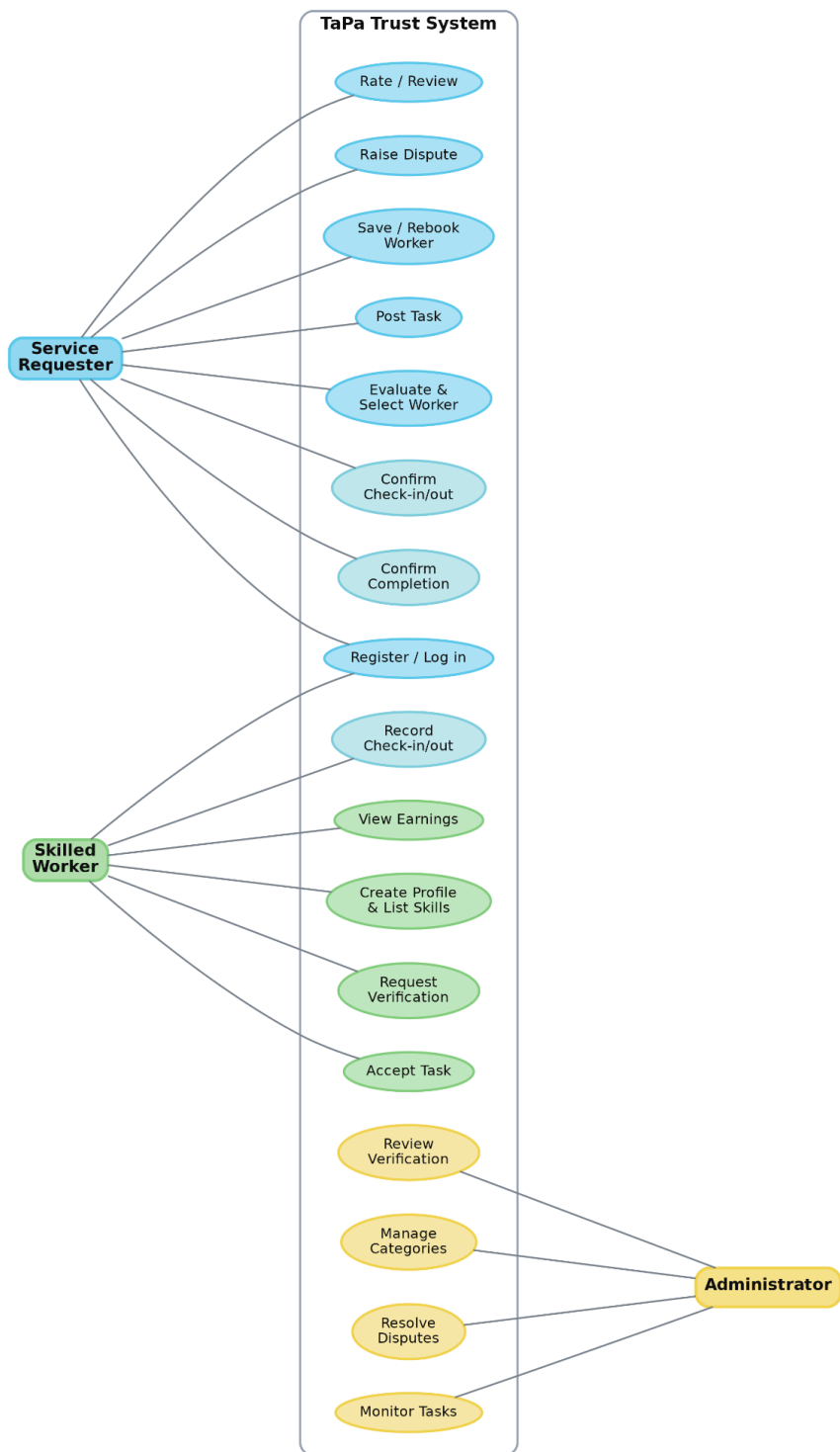


Figure 5: Use Case Diagram

3.8 Sequence Diagram

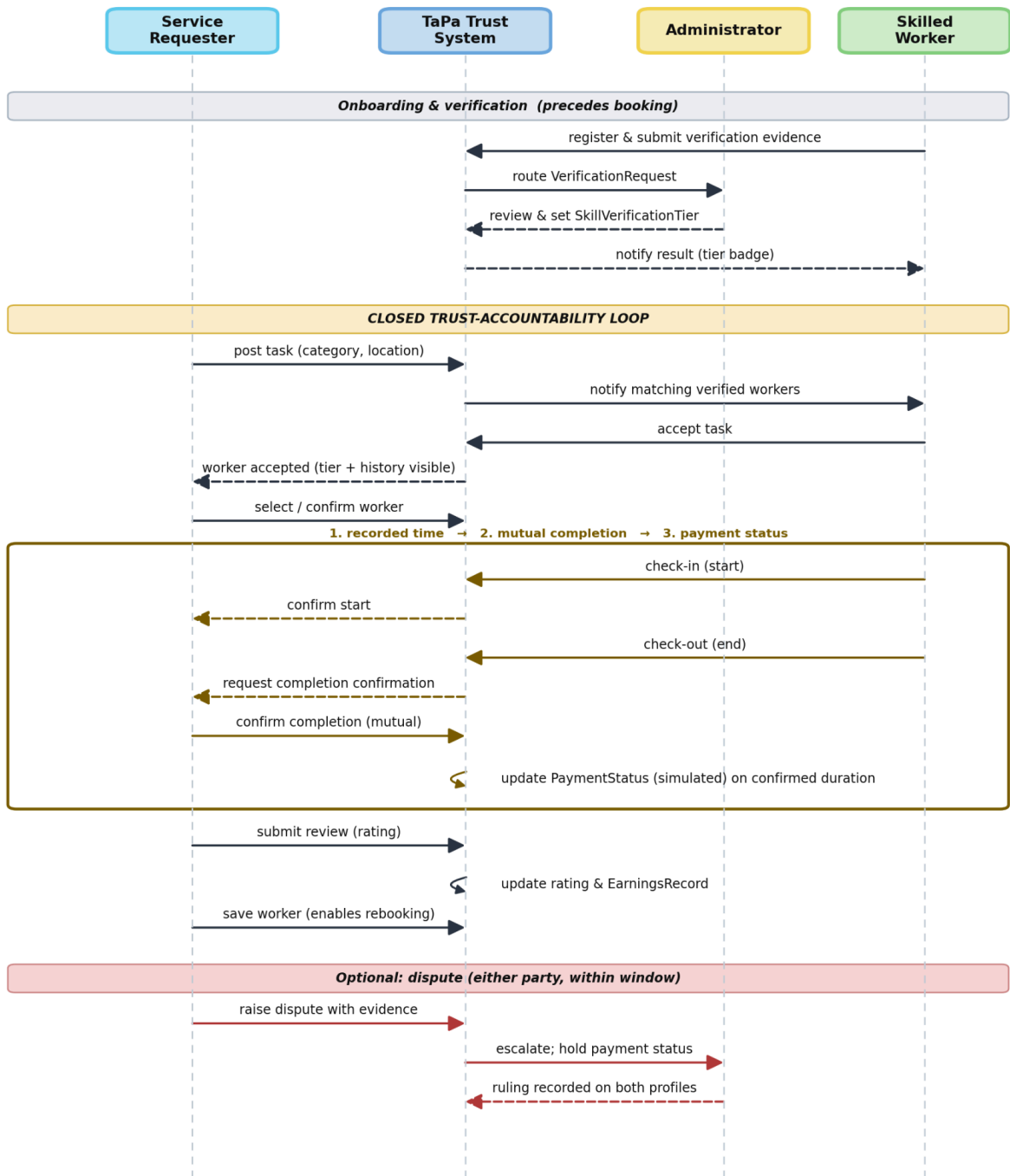


Figure 6: Task Booking Sequence Diagram

Figure 6 traces the full sequence of a trusted task, showing the closed accountability loop in action. It begins with onboarding and verification: the worker submits verification evidence, the system routes a `VerificationRequest`, the administrator reviews it and sets the `SkillVerificationTier`, and the worker is notified of the result. The main loop then runs — the

requester posts a task and selects a verified worker; the worker checks in and the requester confirms the start; the worker checks out and the requester confirms completion; the system updates the simulated PaymentStatus based on the confirmed duration; and the requester submits a review and saves the worker for rebooking.

The ordering is intentional rather than incidental. Each step produces evidence that the next step depends on: verification must precede selection so that only trusted workers are matched; check-in/out must be mutually confirmed before payment status can advance, so that the amount owed rests on a time both parties agreed to rather than on one party's claim; and the review and rebooking steps close the loop by feeding the outcome back into the worker's visible history. This mutual-confirmation design is what makes the record defensible if it is later contested. The optional dispute path (shown in red) allows either party to raise a dispute, after which the administrator holds payment status and records a ruling on both profiles. The diagram therefore visualises the project's central contribution: the way verified identity, recorded time, mutual completion, and payment status validate one another in sequence.

3.9 Development Tools

Table 7: Development tools

Tool / technology	Purpose
Figma	UI/UX design and prototype planning
React.js	Web frontend for requesters, workers, and administrators
Node.js and Express.js	Backend API development
PostgreSQL	Relational database for users, workers, tasks, bookings, reviews, verification, check-ins, earnings, disputes
JWT	Authentication and role-based access control
Bcrypt	Password hashing
React Hook Form / Zod	Client-side form validation

Tool / technology	Purpose
Chart.js or Recharts	Worker earnings dashboard charts (Tier 2/3)
Cloudinary or local demo storage	Profile and work-example image uploads
Postman	API testing
GitHub	Version control and project backup
Render/Vercel	Backend and Frontend deployment

This technology stack is selected because it supports fast web-based prototype development, structured backend APIs, relational data storage, and affordable online deployment within the capstone timeline. A web-based prototype is suitable for the first version of TaPa because users and workers can access it through a browser without installing a mobile application. Future work may extend the platform into a mobile application after the web prototype has been tested and validated.

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